

# Feature Guide & Test Script

What the system does, and how to check that it does it

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Written for anyone — no technical background needed

Automated Pre- and Post-Rental Passenger Car Damage Inspection

MSc Dissertation Prototype — APIIT Sri Lanka

Stefania Crishani · CB016792

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## Before you begin

This booklet has two halves. The first describes **what the system can do**, in plain language. The second is a **test script**: a numbered list of things to try, what should happen each time, and a box to tick. Work through it in order and you will have exercised every feature.

You do not need to understand how it works to use this booklet. You need only:

- A web browser.
- The system already running — someone should have started it for you. If the address below does not load, ask them; that part is covered in the User Manual.
- About 20 minutes.

| What     | Value                 |
|----------|-----------------------|
| Address  | http://localhost:4200 |
| Email    | admin@drivetime.lk    |
| Password | ChangeMe123!          |

**You cannot break anything.** This is a practice system with made-up vehicles and rentals. Add, photograph and delete whatever you like. If it ever gets into a state you dislike, whoever set it up can reset it in one command.

## 1. What problem this solves

When a rental car goes out and comes back, someone walks around it looking for damage. That takes up to fifteen minutes, depends on the light, the inspector's eyesight and how long their shift has been, and produces notes that are hard to compare later. When a customer disputes a charge, there is often no clear record of what the car looked like when it left.

This system does three things about that:

- **It looks at your photographs and names the damage it sees** — dents, scratches, cracks, shattered glass, broken lamps, flat tyres.
- **It compares the car before and after the rental** and tells you which damage is new and which was already there.
- **It produces a PDF report** you can put in the rental file or hand to a customer.

**It assists; it does not decide.** Every finding is meant to be confirmed by a person before it reaches a customer. The system is a second pair of eyes that never gets tired and always writes things down — not a replacement for yours.

## 2. Everything the system can do

The five items down the left-hand side of the screen are the whole system.

| Where                                                                                                                                                                                                                                                                                                                                  | What it is for                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                     |
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| Where                                                                                                                                                                                                                                                                                                                                | What it is for                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                     |
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## 2.1 Recording an inspection

- **Two kinds of inspection.** *Pre-rental* is the condition at handover. *Post-rental* is the condition on return.
- **A rental reference ties them together.** Type the same reference on both and the system knows they are two halves of one rental. This is the single most important field on the form.
- **Photographs are dragged in**, several at once. You tell it which angle each batch was taken from, so the report says 'Driver side' rather than 'photo 3'.
- **Analysis happens automatically** when you upload. Results appear in a second or two per photograph.
- **Odometer, location and notes** are optional, and go on the report.

## 2.2 Reading the findings

Each photograph gets a row per damage type with a coloured bar. Two things on that bar matter:

- **The coloured bar** is how confident the system is.

- **The small grey tick mark** is the level that type of damage has to reach before it counts. Each type has its own mark, because some are easier to spot than others. Only a bar that reaches its own tick is reported as damage.

Damage that is reported appears in full colour and is listed at the top of the inspection. Everything else is greyed out. Hover over any damage name and it will tell you how reliable the system is at spotting that particular type.

**Severity** (minor / moderate / severe) is a rough hint based on how confident the system is — not a measurement of how big the damage is. Treat it as a nudge about where to look first, nothing more.

## 2.3 The comparison — what this is really for

Choose a vehicle on the **Pre / post** screen and it sorts damage into three:

| Outcome                       | Plain meaning                                                                                 | Charge the customer?                        |
|-------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>New this rental</b>        | Not there at handover; there on return.                                                       | <b>Yes</b> — it happened during the rental. |
| <b>Pre-existing</b>           | Was already there when the car went out.                                                      | No — not their doing.                       |
| <b>Not detected on return</b> | Seen at handover but not on return. Usually repaired, or photographed from a different angle. | No.                                         |

## 2.4 Reports

- **Inspection report** — one vehicle, one check. The record, the vehicle, the damage found with confidence figures, and a page of the photographs.
- **Comparison report** — before and after side by side, with new damage highlighted in red. This is the one to attach to a disputed charge.

Every photograph carries a long code called a *fingerprint*, recorded the moment it is uploaded. If anyone ever asks whether the picture in the report is really the picture that was examined, that code is the proof.

## 2.5 Accounts

| Account              | What they can do                                          |
|----------------------|-----------------------------------------------------------|
| <b>Inspector</b>     | Run inspections, view everything, edit their own records. |
| <b>Administrator</b> | All of that, plus manage users and edit anyone's records. |

## 3. What it is *not* good at

Knowing the weak spots is part of using it properly. None of this is hidden in the system — it is all on the **Model** page.

| Limitation                                               | What it means for you                                                                                                               |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cracks are often missed</b>                           | It finds fewer than half of them. Cracks are thin and low-contrast. Check for cracks yourself; do not rely on the system for these. |
| <b>It sometimes adds a damage type that is not there</b> | Usually 'scratch' alongside a correct finding. Look at the photograph before acting on any single finding.                          |
| <b>It only sees what you photograph</b>                  | An unphotographed panel is not an undamaged panel. Photograph everything you would check by hand.                                   |
| <b>It does not measure size or cost</b>                  | It says 'there is a dent', not 'the dent is 4cm' or 'this costs £200'.                                                              |
| <b>It does not read number plates or identify people</b> | By design. It records vehicle condition only.                                                                                       |

## 4. Test script

Work through these in order. Each one says what to do and what should happen. Tick the box, and write a note if something looks wrong. Tests 1 to 8 cover normal use; 9 to 12 check that the system is honest about its own accuracy.

Tests 9 to 12 are the important ones if you are deciding whether to trust this system. Anyone can make a demo look good; these check whether it tells you the truth about itself.

### Part A — Everyday use

#### Test 1 — Signing in

Do this:

1. Open `http://localhost:4200` in your browser.
2. Click the **admin@drivetime.lk** row under *Seeded accounts* — it fills the form for you.
3. Click **Sign in**.

You should see:

- The dashboard appears, with your name at the top right.
- Five items down the left: Dashboard, Inspections, Fleet, Pre / post, Model.

Result: ☐ Works ☐ Does not work Notes: \_\_\_\_\_

#### Test 2 — The dashboard tells you something useful

Do this:

1. Look at the four boxes across the top.
2. Look at **Damage detected across the fleet**.
3. Click any row under **Recent inspections**.

You should see:

- The four boxes show counts, not zeroes or blanks.
- The damage chart lists damage types with coloured bars.
- Clicking a row opens that inspection.

Result: ☐ Works ☐ Does not work Notes: \_\_\_\_\_

#### Test 3 — Adding a vehicle

Do this:

1. Click **Fleet**, then **+ Add vehicle**.
2. Registration test-0001, make Toyota, model Corolla.
3. Click **Save vehicle**.

You should see:

- The vehicle appears in the list as **TEST-0001** — it is tidied to capitals automatically.
- Trying to add the same registration again is refused.

Result: ☐ Works ☐ Does not work Notes: \_\_\_\_\_

### Test 4 — Recording a handover (pre-rental)

**Do this:**

1. Click **+ New inspection**.
2. Choose **Pre-rental**, pick **TEST-0001**.
3. Rental reference: TEST-100. Remember it.
4. Click **Create and add photographs**.
5. Choose a capture angle, then drag in two or three car photographs and click **Upload and analyse**.

**You should see:**

- Each photograph appears with a list of damage types and coloured bars.
- Anything found is listed at the top under **Damage detected**.
- The status changes to **Completed**.

**Result:** ☐ Works ☐ Does not work Notes: \_\_\_\_\_

### Test 5 — Recording the return (post-rental)

**Do this:**

1. Click **+ New inspection** again.
2. Choose **Post-rental**, the same vehicle **TEST-0001**.
3. Rental reference: TEST-100 — exactly the same as before.
4. Upload some different photographs, ideally of a more damaged car.

**You should see:**

- A second inspection is created for the same vehicle.
- Both now appear in the vehicle's history.

**Result:** ☐ Works ☐ Does not work Notes: \_\_\_\_\_

### Test 6 — The comparison — the main event

**Do this:**

1. Click **Pre / post** in the sidebar.
2. Choose **TEST-0001**, type TEST-100, click **Compare**.

**You should see:**

- A verdict panel says whether new damage was found.
- A **Damage delta** table lists each damage type as *New this rental*, *Pre-existing* or *Not detected on return*.
- New damage rows are highlighted in red.
- Both sets of photographs appear side by side underneath.

**Result:** ☐ Works ☐ Does not work Notes: \_\_\_\_\_

### ***Test 7 — Exporting a report***

**Do this:**

1. Open any inspection and click **Export PDF**.
2. Go back to **Pre / post** and click **Export comparison PDF**.
3. Open both files.

**You should see:**

- Both download and open as proper PDFs.
- They contain the vehicle, the inspector, the date, the damage found, and the photographs.
- The comparison report highlights new damage.

**Result:** ☐ Works ☐ Does not work Notes: \_\_\_\_\_

### ***Test 8 — The history is kept***

**Do this:**

1. Click **Fleet**, then open **TEST-0001**.

**You should see:**

- Both inspections are listed with dates, types and findings.
- **Condition summary** reflects the most recent one.

**Result:** ☐ Works ☐ Does not work Notes: \_\_\_\_\_

## Part B — Is it telling you the truth?

These four tests are what separate a system you can rely on from one that merely looks impressive.

### Test 9 — *It publishes its own accuracy, including its failures*

Do this:

1. Click **Model** in the sidebar.
2. Read the boxes across the top.
3. Scroll to **Spot check** and click **Mistakes only**.

You should see:

- It reports being tested on 643 photographs it had never seen before.
- **Damaged vs clean** is about 0.9922, and **Macro F1** about 0.7799.
- The per-class table shows one damage type scoring clearly worse than the others (crack) — it does not hide this.
- **Mistakes only** lists photographs it got wrong, beside the correct answer. A system that could not show you its mistakes would not be trustworthy.

Result: ☐ Works ☐ Does not work Notes: \_\_\_\_\_

### Test 10 — *It does not cry wolf on an undamaged car*

Do this:

1. On the **Model** page, find the **False alarms** box.
2. Then: photograph a car with no damage — any clean car, your own phone is fine.
3. Create an inspection and upload that photograph.

You should see:

- The **False alarms** box shows about **0.0%**, measured over 269 undamaged cars, and is green.
- Your own clean car photograph comes back with **no damage reported**.
- **If it reports damage on a clean car, stop and tell whoever set this up.** That is the most serious failure this system can have — it would mean accusing customers of damage they did not cause.

Result: ☐ Works ☐ Does not work Notes: \_\_\_\_\_

### Test 11 — *It gives the same answer twice*

Do this:

1. Upload the same photograph to two different inspections.
2. Compare the percentages on each.

You should see:

- The percentages are identical, digit for digit.
- If the same photograph gave different answers on different days, no report based on it could be defended in a dispute.

Result: ☐ Works ☐ Does not work Notes: \_\_\_\_\_

**Test 12 — It gets known photographs right****Do this:**

1. Ask whoever set the system up for the **verification pack** — a folder of photographs with a written answer key.
2. Create an inspection and upload all of them.
3. Compare each result against the answer key.

**You should see:**

- It finds the real damage in nearly every photograph.
- Where it is wrong, it has usually *added* an extra damage type rather than missed the real one.
- It is noticeably weaker on cracks — expected, and stated on the Model page.

**Result:** ☐ Works ☐ Does not work Notes: \_\_\_\_\_

## 5. Words you will see, in plain English

| Term                         | What it means                                                                                                                           |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pre-rental</b>            | The check done when the car goes out to a customer.                                                                                     |
| <b>Post-rental</b>           | The check done when it comes back.                                                                                                      |
| <b>Rental reference</b>      | A code you invent that links the two checks of one rental. Use the same one on both.                                                    |
| <b>Confidence</b>            | How sure the system is, as a percentage. Higher means more sure.                                                                        |
| <b>Threshold</b>             | The level confidence must reach before something is reported as damage. Each damage type has its own, shown as a small tick on the bar. |
| <b>False alarm</b>           | Reporting damage on a car that has none. The failure that matters most.                                                                 |
| <b>Precision</b>             | Of the damage it reported, how much was really there. Low precision means false accusations.                                            |
| <b>Recall</b>                | Of the damage that was really there, how much it found. Low recall means missed damage.                                                 |
| <b>F1</b>                    | One number balancing precision and recall. 1.0 is perfect; below about 0.5 is unreliable on its own.                                    |
| <b>Held-out photographs</b>  | Pictures deliberately kept away from the system while it was learning, so testing it on them is a fair exam rather than a memory test.  |
| <b>Severity hint</b>         | A rough minor/moderate/severe label based on confidence. Not a measurement of how big the damage is.                                    |
| <b>Fingerprint (SHA-256)</b> | A code unique to each photograph, recorded on upload. Proves the picture in the report is the picture that was examined.                |

## 6. If something goes wrong

| What you see                                          | What to do                                                                                                 |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| The page will not load, or signing in fails           | The system is not running. Ask whoever set it up to start it.                                              |
| An amber banner saying <i>Placeholder predictions</i> | No trained model is loaded, so all findings are meaningless. Do not use any result until this is resolved. |
| It reports damage on a clearly undamaged car          | Stop and report it. See Test 10.                                                                           |
| A photograph is rejected                              | It must be a JPEG, PNG or WebP image under 15 MB. Photos straight from a phone are fine.                   |
| You are signed out unexpectedly                       | Sessions last 8 hours. Sign in again.                                                                      |
| You want to start over                                | Ask whoever set it up; it is a single command and nothing real is lost.                                    |

**The one rule worth remembering.** The system is a second pair of eyes that never gets tired and always writes things down. It is not a judge. Look at the photograph before you act on any finding, and never charge a customer on the strength of a number alone.